

Denis Michira

Software Engineer · Backend Developer · Full Stack Developer

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience building scalable backend systems and enterprise applications across banking, financial services, insurance, HR, and ticketing domains. Specialises in C#, ASP.NET Core (.NET 10), and Clean Architecture with a strong focus on security, audit-driven development, and API-first systems. Proven track record integrating core banking systems (IMAL), designing microservice-based JWT/RBAC authentication frameworks, delivering high-performance PostgreSQL solutions using stored functions and ADO.NET, and building rich React/Redux frontends. Experienced in Azure, Docker, CI/CD, Excel/PDF report generation, and database migration and testing.

TECHNICAL SKILLS

Languages	C#, Java, JavaScript (ES6+), TypeScript, SQL
Backend	ASP.NET Core Web API (.NET 6–10), RESTful APIs, GraphQL, SOAP Web Services, Microservices
Frontend	React, Redux Toolkit, TypeScript, HTML5, CSS3
Databases	PostgreSQL (stored functions, JSONB), MSSQL, MySQL
Data Access	ADO.NET, Entity Framework Core
Architecture	Clean Architecture, SOLID Principles, Repository Pattern, Service Layer Pattern
Security	JWT Authentication, Refresh Tokens, Role-Based Access Control (RBAC), Permission Middleware
Testing	Unit Testing (xUnit), White-box Testing, Database Migration Testing & Data Validation
Cloud & DevOps	Azure (App Services, Blob Storage), Docker, GitHub Actions (CI/CD)
Reporting	ClosedXML (Excel), PDF generation, Leaderboard & timesheet reports
Integrations	M-PESA, IMAL Core Banking (XML/JSON gateway), Third-party banking systems
Other	NLog (structured logging), Audit Trail (DB), OpenAPI/Swagger, Git

PROFESSIONAL EXPERIENCE

Smartcode Technologies — Nairobi, Kenya

Senior Software Engineer

January 2023 – Present

- Wrote database migration scripts converting schemas and data from MySQL to PostgreSQL, carrying out migration testing and record-level validation to confirm data integrity across environments.
- Applied white-box testing practices to critical service and repository logic, writing xUnit test cases around stored-function calls and permission checks to catch regressions before release.
- Architected a Ticketing & Timesheet Management System using C#, ASP.NET Core (.NET 10) and PostgreSQL following Clean Architecture — split into Authentication API, Main API, and ASP.NET Core MVC UI.
- Built a dedicated Authentication API with JWT token issuance, refresh token rotation, and a granular role and permission system; all Main API endpoints are validated against this service before access is granted.

- Developed the MVC UI layer with a custom middleware that dynamically renders only menus and views permitted by the user's role, calling every function through the API layer.
- Implemented the entire data access layer using ADO.NET with PostgreSQL stored functions — zero raw SQL in application code — ensuring consistent, reusable, and auditable database logic.
- Built a reporting module generating timesheet and ticket-handling leaderboards by period, downloadable in Excel (ClosedXML) and PDF formats.
- Implemented NLog structured logging and a database audit trail across all three projects for full traceability.
- Designed an ERP Banking API Gateway that accepts JSON from all bank channels and translates requests to XML for the IMAL core banking system, returning uniform structured responses — reducing direct core exposure and improving security.
- Designed and built 15+ RESTful and GraphQL APIs using ASP.NET Core, applying scalable architecture patterns and microservices design.
- Integrated M-PESA and third-party banking systems, improving automation and transaction processing efficiency.
- Improved database performance by 40% through query optimisation, indexing strategies, and stored function design.
- Migrated applications to Azure cloud infrastructure; maintained CI/CD pipelines and containerised services with Docker.

Smartcode Technologies — Nairobi, Kenya

Junior Software Engineer

January 2022 – December 2022

- Built a Property Management back-office UI using React and Redux Toolkit, managing global application state across the entire interface.
- Designed and implemented a suite of reusable table and UI components in React used consistently across modules.
- Developed backend services and APIs using ASP.NET Core, and consumed REST APIs across multiple services.
- Wrote SQL queries and supported database design for application modules.

Smartcode Technologies — Nairobi, Kenya

Software Engineering Intern / Attachment

August 2021 – December 2021

- Contributed to backend feature development and bug fixes across live projects.
- Gained hands-on experience with REST APIs, SQL, and Git workflows.
- Participated in Agile ceremonies, sprint planning, and code reviews.

KEY PROJECT HIGHLIGHTS

Ticketing & Timesheet Management System — ASP.NET Core (.NET 10), PostgreSQL, ADO.NET, JWT, React

- Three-tier architecture: Authentication API (JWT + RBAC), Main API (permission-secured endpoints), MVC UI (role-aware middleware navigation).
- Full Clean Architecture: APIs, Domain, Application, Infrastructure (single DB communication class), Shared — ensuring strict separation of concerns.
- All database logic in PostgreSQL stored functions via ADO.NET; no ad-hoc SQL in application code.

- Excel and PDF leaderboard reports by agent and time period. NLog logging + full database audit trail.

ERP Banking API Gateway — ASP.NET Core, IMAL Core Banking, XML/JSON, Azure

- Centralised JSON-to-XML translation layer connecting all bank channels to the IMAL core banking system.
- Enforces a single secure channel to the core, reducing attack surface and standardising responses across all consumers.
- Enabled seamless onboarding of new bank channels without modifying core banking integration logic.

Property Management Back-Office UI — React, Redux Toolkit, TypeScript

- Built a fully featured back-office management interface with Redux Toolkit for centralised state management.
- Delivered a reusable component library of table and form components used across all modules of the system.

EDUCATION

KCA University — Nairobi, Kenya

Bachelor of Science – Information Technology

September 2017 – April 2021